

EXPERIMENT 7: POWER BI VISUALIZATIONS

Data Analytics & Business Intelligence Laboratory Manual

Subject: Power BI Data Visualization

Software Required: Power BI Desktop

Experiment No: 07

File Extension: .pbix

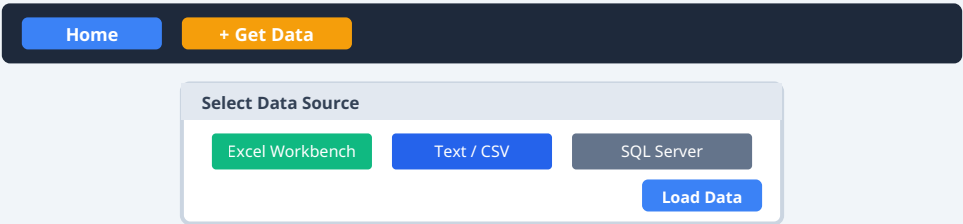
1. Aim

To import a dataset into Power BI Desktop and create fundamental visualizations including a Clustered Column Chart, Clustered Bar Chart, Pie Chart, Line Chart, Card metric, and an interactive Category Slicer to build a functional analytical report dashboard.

2. Step-by-Step Procedure & Visual Guide

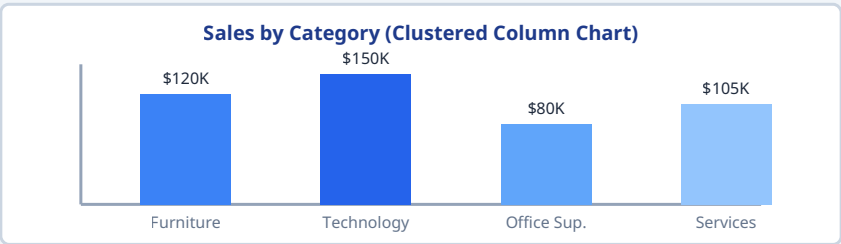
Step 1 & Step 2: Launch Power BI Desktop & Import Dataset

Open Power BI Desktop. Navigate to **Home** → **Get Data** → **Text/CSV** (or Excel). Select the dataset provided (e.g., Sales_Data.csv) and click **Load** in the data preview dialog.



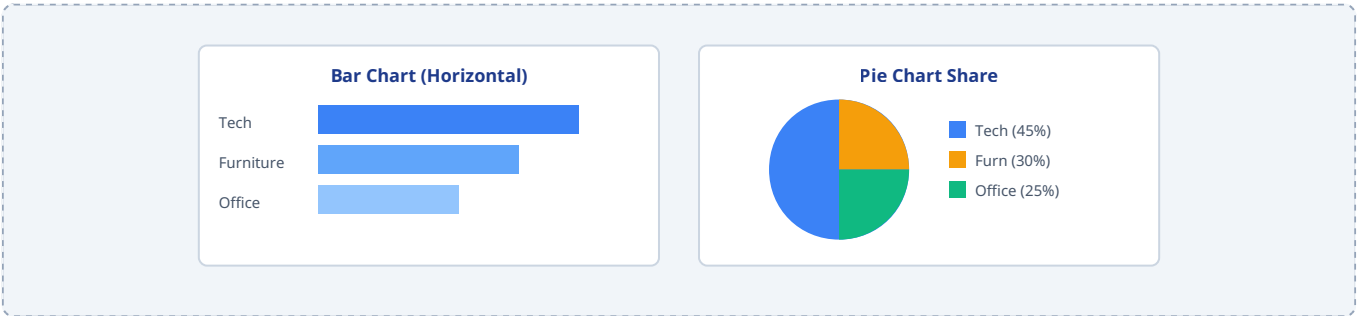
Step 3 & Step 4: Interface Overview & Clustered Column Chart

Switch to **Report View**. From the *Visualizations Pane*, select **Clustered Column Chart**. Drag **Category** into the **X-axis** field and **Sales** into the **Y-axis** field.



Step 5 & Step 6: Clustered Bar Chart & Pie Chart Visuals

Create a **Clustered Bar Chart** (Category → Y-axis, Sales → X-axis) to compare values horizontally. Then add a **Pie Chart** by setting Category to *Legend* and Sales to *Values* to show market share proportions.

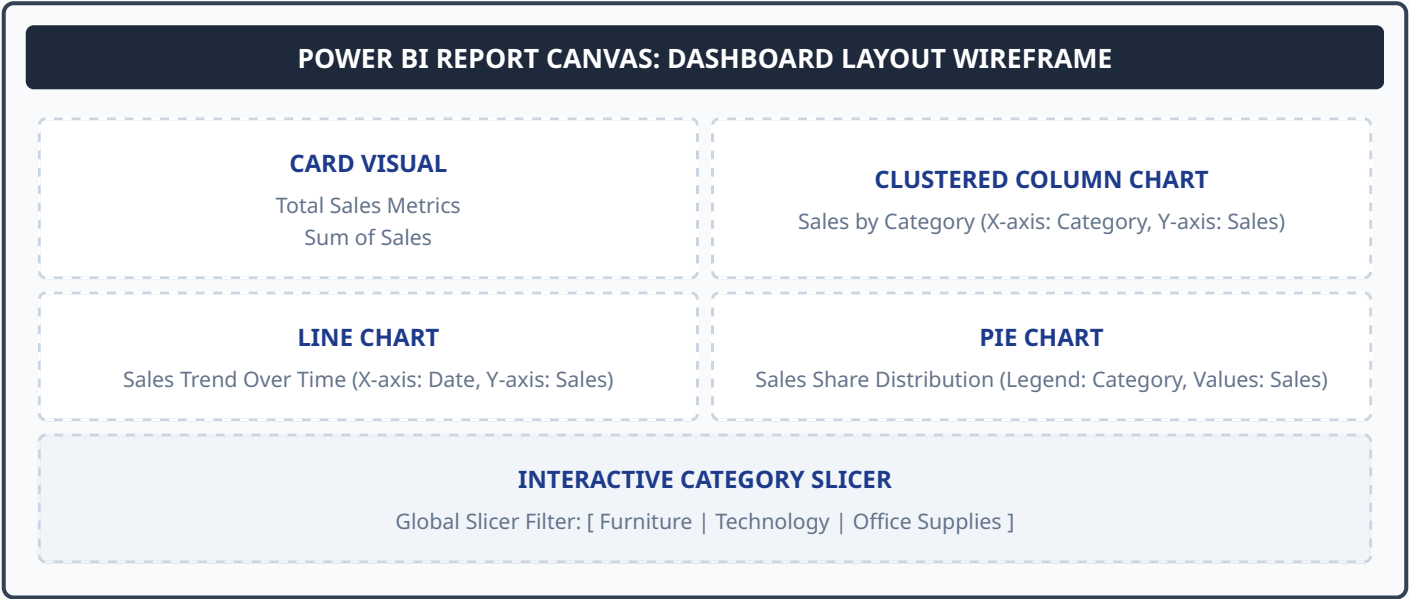


Step 7, 8 & 10: Line Chart, Card KPI & Interactive Slicer

Add a **Line Chart** (Date → X-axis, Sales → Y-axis) for trend tracking. Insert a **Card** visual for aggregate revenue metrics, and place a **Slicer** on the canvas to enable interactive filtering across all visuals.



3. Final Report Canvas Architecture (Step 11)



4. Viva Voce Questions & Answers

Q1. What is Power BI?

Power BI is a Business Intelligence and Data Visualization platform developed by Microsoft that transforms raw data into interactive reports, KPI dashboards, and actionable insights.

Q2. What is the difference between Power BI Desktop and Power BI Service?

Power BI Desktop is a free local application used for data cleaning, modeling, building reports, and DAX calculations. Power BI Service is a cloud-based SaaS platform used for publishing, sharing reports, and building organizational dashboards.

Q3. What is a visualization?

A visualization is a graphical representation of data (e.g., charts, graphs, maps, cards) designed to make complex data trends, patterns, and anomalies visually understandable.

Q4. What is a slicer in Power BI?

A slicer is an interactive visual filter placed directly on the report canvas that enables users to dynamically segment and filter all other visuals on the report page.

Q5. What is the purpose of a Card visual?

A Card visual is used to highlight a single important Key Performance Indicator (KPI) numeric value, such as Total Sales or Total Profit, in a bold summary format.

Q6. What is the difference between a bar chart and a column chart?

A Column Chart presents data vertically, which is suitable for temporal sequences or short category labels. A Bar Chart presents data horizontally, which works best when category labels are long or when comparing many items.

Q7. When is a line chart useful?

A line chart is optimal for visualizing continuous time-series data to analyze trends, seasonal patterns, or growth trajectory over time (e.g., daily, monthly, or yearly sales).

Q8. What is data aggregation in Power BI?

Data aggregation is the mathematical process of combining multiple row values into a single consolidated summary metric, such as SUM, AVERAGE, COUNT, MIN, or MAX.

Q9. What is the difference between a report and a dashboard?

A Report is a multi-visual, highly interactive single or multi-page layout built on a single dataset. A Dashboard is a single-page canvas in Power BI Service that pins summary visual tiles from multiple underlying reports and datasets.

Q10. What is a .pbix file?

A .pbix file is the default proprietary file format for Power BI Desktop documents, containing the imported data model, queries, visual layouts, and DAX expressions in a single file package.

5. Result

Conclusion: The sample dataset was successfully imported into Power BI Desktop. Fundamental visuals including Column Chart, Bar Chart, Pie Chart, Line Chart, Card KPI, and interactive Slicers were configured, formatted, and verified against faculty test cases. The project file was saved as `EXP7_Basic_Visualizations.pbix`.